**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 182 · Recursion: call stack and tracing

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. For Factorial(3), list the successive input values before any returns occur.

Q2. What is the difference between a function calling another function and returning a value to its caller?

Q3. A stack contains bottom to top P, Q, R. After one pop, what is at the top?

Q4. A function Square(x) returns x*x. Find Square(2)+Square(3), showing both calls.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

stack frame — An entry containing the state for one active function call.

unwind — Return from calls in reverse order after a base case is reached.

local variable — A variable owned by a particular invocation.

call stack — Last-in, first-out storage for active calls.

base case — A condition that ends recursion.

## New learning

Build the stack — SumTo(4) waits for SumTo(3), which waits for SumTo(2), which waits for SumTo(1).

Separate the local values — Each stack frame holds its own n: 4, 3, 2 or 1. The values do not overwrite one another.

Unwind from the top — SumTo(1) returns 1; SumTo(2) returns 3; SumTo(3) returns 6; SumTo(4) returns 10.

## Worked example

Draw the call-stack frames for SumTo(3). Which call returns first?

1. Push — The frames are SumTo(3), SumTo(2), then SumTo(1), with SumTo(1) at the top.

2. Stop — SumTo(1) is the base case; it returns 1 first.

3. Unwind — SumTo(2)=2+1=3, followed by SumTo(3)=3+3=6.

# We Do · Practise with guidance

Trace SumTo(5). Label the top of the stack immediately before the first return.

5 → 4 → 3 → 2 → 1

Predict the next step before revealing the explanation.

Call sequence

SumTo(5) → SumTo(4) → SumTo(3) → SumTo(2) → SumTo(1).

Top frame

SumTo(1) is on top immediately before the first return.

Returns

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1, 3, 6, 10, 15. SumTo(5) returns 15.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

List the call-stack frames from bottom to top for SumTo(4).

### Question 2

Trace the returned values for SumTo(6).

### Question 3

A pupil writes that SumTo(4) returns 4+3+2+1 before SumTo(1) has executed. Explain the order error.

## Exit Ticket · Answer before checking

Every recursive call uses the same local-variable storage.







## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. 3, 2, 1, 0. Each call reduces n by one until the base case.

Q2. A call transfers control into a function; return finishes that invocation and gives its result to the caller.

Q3. Q. The topmost R is removed first.

Q4. Square(2)=4 and Square(3)=9; the total is 13.

### You Do pages · Independent questions

Q1. SumTo(4), SumTo(3), SumTo(2), SumTo(1). SumTo(1) is the top frame.

Q2. 1, 3, 6, 10, 15, 21 in the order calls return; the outer function returns 21.

Q3. The expression describes the final arithmetic but not execution order. The deepest call SumTo(1) must return first; only then can waiting calls compute their sums.

### Exit Ticket · Model answer

False. Each active call has its own stack frame and local values.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**